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AI governance around the world

Country profile: Singapore

Ana Alania

About The Alan Turing Institute

The Alan Turing Institute is the UK's national institute for data science and artificial intelligence.

Authors

Ana Alania is a Research Manager in AI Governance at the Alan Turing Institute.

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About the project

Introduction

The international AI governance landscape continues to evolve as jurisdictions around the world balance the need to navigate increased competition with earlier commitments to advancing international alignment and cooperation on AI governance and safety. Many jurisdictions have recognised the strategic implications of AI, both for economic prosperity and national security, and accelerated investment in domestic AI infrastructure has begun to engender a more competitive international environment. However, the role of international trade and global cooperation remains crucial in realising the economic benefits of AI while effectively managing the technology's risks.

As high-level AI principles are translated into concrete policies and distinct regulatory frameworks, jurisdictions are managing the tensions between competitive and cooperative priorities. Meaningful progress towards effective international AI governance will require a clear and detailed understanding of how different countries are approaching AI governance in practice. Tracking primary source policy initiatives over the past decade, the *AI governance around the world* project sheds light on shifting national priorities on AI, dynamics of cooperation and competition, and potential areas of international alignment.

Project aims

The *AI governance around the world* project seeks to map this evolving landscape through a series of country profiles based on a consistent framework. Each profile provides a descriptive overview of a jurisdiction's approach to AI regulation and standardisation, highlighting the high-level aims and principles, definitions of relevant technologies, key policy initiatives and the main features of the respective standardisation systems. Drawing exclusively on primary sources, the country profiles analyse legal and policy instruments, national strategies, standardisation initiatives, and public investments that reflect the varied and evolving approaches that different jurisdictions are taking to AI governance.

This series offers a foundation for comparative analysis and future work on global regulatory interoperability without commenting on the efficacy of the specific governance models being adopted at the jurisdictional level. As more profiles are developed, the project aims to contribute to international understanding of where alignment is emerging and where deeper coordination may be needed.

Singapore

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Executive summary

Singapore's evolving approach to AI governance prioritises innovation and the strategic deployment of AI technologies in sectors with high socio-economic impact. While the government has not introduced AI-specific regulation, several coordinated initiatives have been launched to encourage the responsible development and use of AI technologies across industry. This includes sector-specific voluntary guidelines introduced by the Monetary Authority of Singapore, as well as horizontal non-binding frameworks such as the Model AI Governance Framework. Singapore's non-prescriptive approach to governing AI is geared towards providing practical tools and guidelines to support businesses and strengthening the country's AI assurance ecosystem through the development of testing frameworks and self-assessment toolkits.

In parallel, Singapore is making significant progress towards AI standardisation through a combination of international and domestic initiatives. A dedicated technical committee for AI under the Singapore Standards Council (SSC) is actively working to shape and adopt international AI standards, while also developing national standards to address domestic industry needs. Singapore's interest in standardisation aligns well with the country's overall pro-innovation approach to AI policy.

Key policy initiatives

Nov 2018 **FEAT Principles in Financial Sector**

Ethical guidelines promoting fairness, ethics, accountability, and transparency in AI use within the financial sector.

Jan 2019 **Model AI Governance Framework**

Voluntary framework to guide responsible AI development and use in Singapore. The second edition was published in Jan 2020.

Nov 2019 **National AI Strategy (NAIS)**

Strategic roadmap identifying priority AI projects and ecosystem enablers. NAIS 2.0 was published in Dec 2023.

May 2022 **AI Verify**

Self-assessment toolkit enabling organisations to test and demonstrate AI system trustworthiness.

Jun 2023 **AI Verify Foundation**

Not-for-profit entity developing open-source tools, benchmarks, and best practices for AI testing globally.

May 2024 **Governance Framework for Generative AI**

Builds on the Model AI Governance Framework to provide specific governance considerations for generative AI.

May 2024 **Singapore AI Safety Institute**

National centre for AI safety science, research, and international cooperation.

Regulatory approach to AI

High-level aims and principles

Singapore's national approach to AI is grounded in the broader vision set out in the government's Smart Nation initiative, launched in November 2014. Framed as a whole-of-government effort, the initiative aims to harness emerging technological solutions to address national challenges and drive digital transformation in key sectors such as health, urban mobility, and finance.[1]

Artificial Intelligence has been identified as one of four frontier technologies critical to realising these ambitions, with the government emphasising its potential to deliver "quantum leaps in productivity and efficiency".[2] The Smart Nation and Digital Government Office has articulated a vision for Singapore to become "a global hub for developing, test-bedding, deploying, and scaling AI solutions" and to use AI as a driver of economic growth in high-impact sectors.[3] While the overall policy approach is undoubtedly pro-innovation, building a trusted environment around AI has been recognised as key to achieving these goals. To this end, a range of horizontal and vertical policy initiatives have been introduced to guide responsible development and use of AI.

These initiatives are largely informed by internationally recognised AI ethics principles, such as those proposed by the EU, OECD, and IEEE. For example, the *Model AI Governance Framework* highlights 12 foundational principles distilled from various international sources but notes not all of them are addressed in the framework. These are accountability, accuracy, auditability, explainability, fairness, human centricity and well-being, human rights alignment, inclusivity, progressiveness, transparency, robustness and security, and sustainability.[4] Most of Singapore's AI-focused policies

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[1] Smart Nation and Digital Government Office., 'Smart Nation: The Way Forward', November 2018, 28, https://www.smartnation.gov.sg/files/publications/smart-nation-strategy-nov2018.pdf.

[2] Smart Nation and Digital Government Office, 'Smart Nation: The Way Forward'.

[3] Smart Nation and Digital Government Office, 'National Artificial Intelligence Strategy - Advancing Our Smart Nation Journey', November 2019, 7, https://www.smartnation.gov.sg/files/publications/national-ai-strategy.pdf.

[4] Info-communications Media Development Authority and Personal Data Protection Commission, 'Model AI Governance Framework - Second Edition', January 2020, https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgmodelaigovframework2.pdf.

aim to assist organisations in translating some of these high-level principles into concrete practices that can be assessed and verified to demonstrate trustworthiness.[5]

Definitions of relevant technologies

Singapore's policy documents use a variety of terms and concepts to refer to AI technologies. The *National AI Strategy* provides a broad definition, referring to AI in terms of "the capability to simulate intelligent, human-like behaviour in computers", which is further contextualised using examples of AI applications like autonomous driving, cancer diagnosis, and fraud detection.[6]

The *Model AI Governance Framework* similarly relates AI to human intelligence and capabilities but expands the definition to include examples of typical outputs derived from AI systems. According to the framework, AI refers to "a set of technologies that seek to simulate human traits such as knowledge, reasoning, problem solving, perception, learning and planning, and, depending on the AI model, produce an output or decision (such as a prediction, recommendation, and/or classification)".[7] Refraining from committing to a single fixed definition, the framework further notes that this "should not be taken to be an authoritative or exhaustive definition".[8]

The Monetary Authority of Singapore defines AI only in the context of automated decision-making, in line with the scope of its *Fairness, Ethics, Accountability and Transparency (FEAT) principles for the use of Artificial Intelligence and Data Analytics (AIDA)* in Singapore's financial sector. According to this document, "AIDA refers to artificial intelligence or data analytics, which are defined as technologies that assist or replace human decision-making".[9]

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[5] Personal Data Protection Commission, 'Singapore's Approach to AI Governance', January 2023, https://www.pdpc.gov.sg/Help-and-Resources/2020/01/Model-AI-Governance-Framework.

[6] Smart Nation and Digital Government Office, 'National Artificial Intelligence Strategy - Advancing Our Smart Nation Journey', 12.

[7] Info-communications Media Development Authority and Personal Data Protection Commission, 'Model AI Governance Framework - Second Edition', 18.

[8] Info-communications Media Development Authority and Personal Data Protection Commission, 18.

[9] Monetary Authority of Singapore, 'Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of Artificial Intelligence and Data Analytics in Singapore's Financial Sector', November 2018, 3, https://www.mas.gov.sg/~/media/MAS/News%20and%20Publications/Monographs%20and%20Information%20Papers/FEAT%20Principles%20Final.pdf.

Key policy initiatives

Horizontal initiatives

Addressing the varied risks and opportunities of AI technologies across sectors, Singapore has enacted multiple non-binding horizontal AI policies and formed an *Advisory Council for AI and Data Ethics*. The *National AI Strategy* outlines the country's vision and strategic direction for AI integration, while policy initiatives such as the *AI Governance Framework* and *AI Verify* focus on providing practical tools for responsibly developing and deploying AI systems.

National Artificial Intelligence Strategy

To set the national agenda for AI, the Smart Nation and Digital Government Office published Singapore's first *National AI Strategy* in November 2019. The strategy identified five "National AI Projects" with a potential to deliver significant social and economic impact for Singapore, including intelligent freight planning, chronic disease prediction and management, and border clearance operations.[10] The second edition of the *National AI Strategy (NAIS 2.0)*, published in December 2023, moves away from projects and introduces three systems underpinned by 10 enablers and 15 concrete actions.[11] Under System 1, titled 'Activity Drivers', NAIS 2.0 sets out plans to strengthen the AI ecosystem within industry, government, and research and promote active collaboration between them. System 2, 'People and Communities', focuses on attracting top-tier AI talent and cultivating a skilled workforce capable of developing, deploying, and using AI systems, models, and algorithms. System 3, 'Infrastructure and Environment', is centred on creating the right conditions for the development of a successful AI value chain, including enabling increased access to compute, building capabilities in data services, and creating a trusted environment.[12]

The updated strategy places a heavier emphasis on creating a 'pro-innovation' regulatory environment for AI that relies on agile regulatory tools, standards, pilots and nurtures domestic Testing, Inspection, and Certification (TIC) sector. Actions outlined in the strategy aim to develop interventions that are 'risk-based, tiered, and adapted for specific vertical sectors and horizontal applications'.[13] Stressing the importance of

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[10] Smart Nation and Digital Government Office, 'National Artificial Intelligence Strategy - Advancing Our Smart Nation Journey'.

[11] Smart Nation Singapore and Ministry of Communications and Information, 'Singapore National AI Strategy 2.0 (NAIS 2.0)', 4 December 2023, https://file.go.gov.sg/nais2023.pdf.

[12] Smart Nation Singapore and Ministry of Communications and Information.

[13] Smart Nation Singapore and Ministry of Communications and Information, 57.

context-specific risk management approaches, the strategy also suggests establishing a central coordinating platform to ensure coherence among regulatory agencies.

The Advisory Council on the Ethical Use of AI and Data

Established in June 2018, the *Advisory Council on the Ethical Use of AI and Data* advises the Singapore government on ethical and legal issues arising from the commercial deployment of AI technologies. It also contributes to the development of voluntary governance frameworks, guidelines, and codes of practice for industry adoption. The members of the Advisory Council are appointed by the Minister for Communications and Information and are, at the time of writing, predominantly representatives of large multinational technology firms such as Microsoft, IBM, Google, and Salesforce.[14]

The Model AI Governance Framework

To address cross-sectoral governance challenges related to AI technologies, Singapore launched the *Model AI Governance Framework* in January 2019. Now in its second edition, the Model Framework translates high-level ethical principles into practical recommendations that private sector organisations can adopt to ensure their AI systems are developed and deployed responsibly.[15] Created jointly by the Personal Data Protection Commission (PDPC) and the Info-communications Media Development Authority (IMDA), the framework is grounded on two high-level principles. First, decisions made by AI should be explainable, transparent, and fair. Second, AI solutions should be human-centric. To help companies operationalise these principles, the framework provides practical guidance in four key areas: (1) internal governance structures and measures, (2) determining the level of human involvement in AI-augmented decision making, (3) operations management, and (4) stakeholder interaction and communication.[16]

The adoption of the *Model AI Governance Framework* is entirely voluntary, with no obligations placed on organisations developing or deploying AI technologies in Singapore. To assist companies wishing to implement it, the PDPC and the IMDA have partnered with the World Economic Forum's Centre for the Fourth Industrial Revolution

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[14] Personal Data Protection Commission, 'Singapore's Approach to AI Governance', January 2023, https://www.pdpc.gov.sg/Help-and-Resources/2020/01/Model-AI-Governance-Framework.

[15] Info-communications Media Development Authority and Personal Data Protection Commission, 'Model AI Governance Framework - Second Edition'.

[16] Info-communications Media Development Authority and Personal Data Protection Commission, 21–62.

to develop the *Implementation and Self-Assessment Guide for Organisations* (ISAGO)[17] and published two volumes of *Compendium of Use Cases*[18] highlighting how organisations have implemented or aligned their internal governance structures to the framework.

AI Verify and AI Verify Foundation

Building on its *Model AI Governance Framework*, IMDA launched *AI Verify* in May 2022 as a Minimum Viable Product (MVP) for international pilot and feedback, describing it as the world's first AI governance testing framework and toolkit.[19] Developed in consultation with the private sector, the toolkit is designed to assist organisations developing or deploying AI systems to demonstrate the trustworthiness of their AI systems through self-conducted technical tests and process checks, aligned with 11 internationally recognised principles.[20]

The MVP initially supported only supervised learning models, such as binary classification and regression, with explainability, robustness, and fairness assessed through technical tests and process checks, while other principles, such as transparency, safety, and data governance, were covered only through process checks.[21] In June 2023, Singapore's Minister for Communications and Information announced the establishment of the *AI Verify Foundation* – a not-for-profit subsidiary of IMDA that would assume responsibility for the continued development of *AI Verify*. The Foundation aims to advance the development of open-source tools, benchmarks, and best practices for AI testing globally, with an emphasis on industry collaboration. Its strategic direction is guided by seven premier members, including IMDA, IBM, Microsoft,

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[17] Info-communications Media Development Authority and Personal Data Protection Commission, 'Companion to the Model AI Governance Framework – Implementation and Self-Assessment Guide for Organizations', January 2020, https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Resource-for-Organisation/AI/SGIsago.pdf.

[18] Info-communications Media Development Authority and Personal Data Protection Commission Singapore, 'Compendium of Use Cases: Practical Illustrations of the Model AI Governance Framework', 2020, https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgaigovusecases.pdf; Info-communications Media Development Authority and Personal Data Protection Commission Singapore, 'Compendium of Use Cases: Practical Illustrations of the Model AI Governance Framework - Volume 2', n.d., https://file.go.gov.sg/ai-gov-use-cases-2.pdf.

[19] Personal Data Protection Commission, 'Singapore's Approach to AI Governance'.

[20] Info-communications Media Development Authority, 'Invitation to Pilot A.I. Verify - AI Governance Testing Framework and Toolkit', May 2022, https://file.go.gov.sg/aiverify.pdf.

[21] AI Verify Foundation, 'Summary Report - Binary Classification Model for Credit Risk', 6 June 2023, https://aiverifyfoundation.sg/downloads/AI_Verify_Sample_Report.pdf.

Google, Salesforce, and Red Hat, with over 120 general members such as Adobe, Meta, and SenseTime.[22] [23]

To complement *AI Verify's* focus on traditional machine learning systems, in May 2025, the *AI Verify Foundation* launched Project Moonshot, an open-source toolkit developed to evaluate risks associated with generative AI. Positioned as one of the world's first large language model (LLM) evaluation toolkits, Project Moonshot integrates benchmarking and red teaming to support developers and compliance teams in testing LLMs and LLM applications.[24]

While encouraging responsible AI practices, *AI Verify Foundation* initiatives remain voluntary, imposing no binding requirements on companies to adopt its frameworks or disclose assessment outcomes. As adoption grows, the Foundation aims to work with global AI system developers and operators to collate industry best practices and develop benchmarks that can contribute to the development of international AI standards.[25] [26]

Model AI Governance Framework for Generative AI

In response to the emerging risks posed by generative AI[27], IMDA and AI Verify Foundation (AIVF) published the *Model AI Governance Framework for Generative AI* in May 2024. The framework proposes "a systematic and balanced approach" to governing generative AI systems, structured around nine interrelated dimensions – Accountability, Data, Trusted Development and Deployment, Incident Reporting, Testing and

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[22] 'Singapore Launches AI Verify Foundation 2023', Infocomm Media Development Authority, 7 June 2023, https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2023/singapore-launches-ai-verify-foundation.
[23] 'Project Moonshot, Powered by AI Verify, and AI Collaborations', Infocomm Media Development Authority, May 2024, https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/factsheets/2024/project-moonshot.
[24] 'Project Moonshot - an LLM Evaluation Toolkit', *AI Verify Foundation*, n.d., accessed 1 July 2025, https://aiverifyfoundation.sg/project-moonshot/.
[25]Info-communications Media Development Authority, 'Invitation to Pilot A.I. Verify - AI Governance Testing Framework and Toolkit', May 2022, https://file.go.gov.sg/aiverify.pdf.

[26] 'Project Moonshot - an LLM Evaluation Toolkit'.
[27] Infocomm Media Development Authority, 'Singapore Proposes Framework to Foster Trusted Generative AI Development', Infocomm Media Development Authority, 16 January 2024, https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2024/public-consult-model-ai-governance-framework-genai.

Assurance, Security, Content Provenance, Safety and Alignment Research and Development, and AI For Public Good.[28]

As with earlier policy documents, the GenAI framework does not set out specific practices or obligations, but instead proposes governance considerations around each dimension that policymakers, industry practitioners and other stakeholders should consider to support the creation of a trusted ecosystem around generative AI. These considerations are informed by earlier work undertaken by IMDA, including a discussion paper on generative AI[29], a catalogue of LLM evaluation benchmarks and methods[30] and a sandbox for generative AI evaluation.[31]

Singapore's AI Safety Institute

Following the AI Seoul Summit in May 2024, a key milestone in international AI safety cooperation, Singapore designated the Digital Trust Centre at Nanyang Technological University (NTU) as its national AI Safety Institute (AISI).[32] The IMDA announced the initiative in alignment with the *Seoul Statement of Intent toward International Cooperation on AI Safety Science*, which called for creating a network of publicly funded safety institutes to advance global AI safety research and testing.[33]

Singapore's AISI has a broad mandate covering technical testing, safety research, policy development, and international collaboration. It aims to address critical gaps in AI safety science, with particular focus on regionally relevant issues like multilingual LLM evaluation.[34] As Singapore's official representative in the International Network of AI

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[28] Info-communications Media Development Authority and AI Verify Foundation, *Model AI Governance Framework for Generative AI: Fostering a Trusted Ecosystem* (2024), 3, https://aiverifyfoundation.sg/wp-content/uploads/2024/06/Model-AI-Governance-Framework-for-Generative-AI-19-June-2024.pdf.
[29] Infocomm Media Development Authority and Aicadium, 'Generative AI: Implications for Trust and Governance', June 2023, https://aiverifyfoundation.sg/downloads/Discussion_Paper.pdf.

[30] Infocomm Media Development Authority and AI Verify Foundation, 'Cataloguing LLM Evaluations', October 2023, https://aiverifyfoundation.sg/downloads/Cataloguing_LLM_Evaluations.pdf.

[31] Infocomm Media Development Authority, 'First of Its Kind Generative AI Evaluation Sandbox for Trusted AI by AI Verify Foundation and IMDA', Infocomm Media Development Authority, Oc 2023, https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2023/generative-ai-evaluation-sandbox.

[32] 'Digital Trust Centre Designated as Singapore's AISI', Infocomm Media Development Authority, 22 May 2024, https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/factsheets/2024/digital-trust-centre.
[33] 'Seoul Statement of Intent toward International Cooperation on AI Safety Science, AI Seoul Summit 2024 (Annex)', GOV.UK, May 2024, https://www.gov.uk/government/publications/seoul-declaration-for-safe-innovative-and-inclusive-ai-ai-seoul-summit-2024/seoul-statement-of-intent-toward-international-cooperation-on-ai-safety-science-ai-seoul-summit-2024-annex.
[34] 'About Us', Singapore AI Safety Institute, 25 January 2025, https://sgaisi.sg/.

Safety Institutes, AISI works closely with global counterparts to develop shared scientific foundations for safely deploying advanced AI models.

Vertical initiatives

Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of Artificial Intelligence and Data Analytics (AIDA) in Singapore's Financial Sector

Leading Singapore's sectoral governance initiatives around AI, in November 2018, the Monetary Authority of Singapore (MAS) published a set of principles for the responsible use of AIDA in the financial sector.[35] The principles consist of four pillars – fairness, ethics, accountability, and transparency (FEAT) – and apply to Singapore's financial institutions that use AIDA systems to assist or replace human decision-making. To help organisations in the practical implementation of these principles, MAS launched the *Veritas* initiative as part of the *National AI Strategy*. The project is delivered across multiple phases in collaboration with some of the largest finance and technology companies such as HSBC, Goldman Sachs, Google, and IBM.[36]

Phase one of the initiative saw the development of two white papers on fairness assessment methodology in credit risk scoring and customer marketing and corresponding case studies gathered from the industry. Phase two concluded in February 2022 and resulted in the development of assessment methodologies for all FEAT principles and Veritas Toolkit version 1. The second iteration of the toolkit was published in June 2023 as part of the phase three outputs, as well as two white papers and case studies detailing guidance on integrating FEAT assessment methodologies.[37]

Similar to the country's horizontal policy initiatives, FEAT principles and the Veritas toolkit act as voluntary guidelines and tools and place no obligations on companies using AIDA systems.

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[35] Monetary Authority of Singapore, 'Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of Artificial Intelligence and Data Analytics in Singapore's Financial Sector'.

[36] Monetary Authority of Singapore, 'Veritas Initiative', October 2023, https://www.mas.gov.sg/schemes-and-initiatives/veritas.

[37] Monetary Authority of Singapore.

Approach to AI standardisation

Main features of the standardisation approach

Enterprise Singapore (ESG), a government agency under the Ministry of Trade and Industry, is responsible for administering the Singapore Standardisation Programme through the industry-led Singapore Standards Council (SSC).[38] As a national standards and accreditation body, ESG also oversees the work of the Singapore Accreditation Council (SAC)[39] and appoints Standards Development Organisations (SDOs) to develop and promote the adoption of standards in specific sectors. ESG has the authority to evaluate new proposals for standards and to edit and approve standards that have gone through the full consensus process with stakeholders.[40]

Current ESG-appointed SDOs include the Institution of Engineers Singapore (IES), Singapore Chemical Industry Council (SCIC), Duke-NUS Centre of Regulatory Excellence (CoRE), Nanyang Polytechnic (NYP), Singapore Manufacturing Federation (SMF), and Energy Research Institute @ Nanyang Technological University (NTU). These SDOs oversee standards committees under their purview and manage associated technical committees and working groups, which are typically comprised of representatives from industry, governmental agencies, trade and consumer associations, professional bodies, and academia. Interestingly, IT standards development is managed by Infocomm Media Development Authority (IMDA), a governmental agency rather than an SDO. IMDA oversees the work of the Information Technology Standards Committee (ITSC).[41]

On a national level, ESG differentiates between three types of standards—Singapore Standards (SS), Technical References (TR), and Workshop Agreements (WA).[42] Singapore Standards are developed by local standards committees under SSC when there are no suitable international or regional standards. TRs are fast-tracked standards

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[38] Enterprise Singapore, 'Develop Standards | Quality & Standards', October 2022, https://www.enterprisesg.gov.sg/quality-standards/standards/for-partners/standards-development.

[39] Singapore Accreditation Council, 'About Singapore Accreditation Council (SAC)', accessed 24 April 2024, https://www.sac-accreditation.gov.sg/about/our-role/.

[40] Singapore Manufacturing Federation, 'Overview | Standards Development Organisation', accessed 24 May 2024, https://www.smf-sdo.org.sg/overview.

[41] Enterprise Singapore.

[42] Enterprise Singapore, 'Standards, Technical References and Workshop Agreements', accessed 24 April 2024, https://www.enterprisesg.gov.sg/grow-your-business/boost-capabilities/quality-and-standards/standards-technical-references-and-workshop-agreements

that are developed in under 12 months to address an urgent industry demand, while WAs can take less than 6 months.[43] For this reason, TRs and WAs are published without going through the full consensus process, skipping the public comment stage. Adoption of Singapore Standards, Technical References, and Workshop Agreements, as well as international standards is voluntary by default but can be made mandatory by a regulatory authority.

Table 1: Overview of National Standards in Singapore

Type	Purpose	Development timeframe	Consensus
Singapore Standards (SS)	Formal, nationally recognised standards	Varies, 12-36 months	Full stakeholder consensus, including public comment
Technical References (TR)	Interim standards for urgent industry needs with three-year validity period	≤ 12 months	Fast-tracked (no public comment)
Workshop Agreements (WA)	Experimental or early-stage practices	≤ 6 months	Limited consensus

National AI-focused standardisation activities

Singapore has established a dedicated technical committee for AI, formed under the SSC-appointed Information Technology Standards Committee (ITSC).[44] The AI Technical Committee (AITC), established in July 2019, is responsible for adopting international AI standards and developing local standards and technical references to meet domestic industry needs that are not yet sufficiently addressed by international initiatives.[45]

[43] Enterprise Singapore, 'Standards: The X Factor - What's the Big Deal about Standards?', November 2022, https://www.enterprisesg.gov.sg/blog/whats-the-big-deal-about-standards

[44] Info-communications Media Development Authority, 'Technical Committees (TCs) / Working Groups (WGs)', February 2021, https://www.imda.gov.sg/-/media/Imda/Files/Regulations-and-Licensing/Licensing/ICT-Standards-and-Quality-of-Service/Industry-Committees-and-Working-Groups/IT-Standards-Committee/Technical-Committees-and-Working-Groups.pdf.

[45] AI Singapore, *AI Standards*, September 2022, https://aisingapore.org/innovation/ai-standards/.

At the time of writing, the AITC has published two horizontal technical references for AI. The first provides guidance on managing potential security risks in the early lifecycle of an AI system, including recommendations on design best practices.[46] The second, TR 139:2025 Artificial Intelligence (AI) Use Cases, presents real-world applications of AI across various sectors in Singapore, offering practical insights into how AI is being deployed nationally.[47] The AITC has also adopted ISO/IEC 42001:2023, Information Technology – Artificial Intelligence – Management System, as a national standard, published as SS ISO/IEC 42001:2024. This is the first international AI standard that the AITC has published as a Singapore Standard. While the national version is identical to the international text, it includes an informative Annex ZA, which cites *AI Verify* as an example of a voluntary testing tool to support alignment with SS ISO/IEC 42001:2024.[48]

Sector-level standardisation initiatives are also well under way in Singapore. Among the standards currently available on the Singapore Standards e-shop are two technical references focused on AI-enabled medical devices developed by the Healthcare Informatics Technical Committee and four technical references on autonomous vehicles developed by the Technical Committee on Automotive. ESG has highlighted the importance of AI standards to "establish trust, reliability and interoperability" among key AI stakeholders and to address gaps in performance, governance, ethics, and cybersecurity issues in the deployment of AI.[49]

Engagement in international AI standardisation

Through the Singapore Standards Council (SSC), the AI Technical Committee represents Singapore as a participating member of ISO/IEC JTC 1/SC 42 on Artificial Intelligence, the key international committee responsible for AI standardisation. Singapore has contributed to several outputs under SC 42, including ISO/IEC TR 24030:2021, Information Technology — Artificial Intelligence — Use Cases, which documents practical examples of AI applications across domains. More broadly, the Singapore Standards Council participates in over 80 ISO technical committees, ensuring close alignment with international standards development.

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[46] *TR 99:2021 Artificial Intelligence (AI) Security – Guidance for Assessing and Defending against AI Security Threats*, n.d., https://www.singaporestandardseshop.sg/Product/SSPdtDetail/553b4562-ef85-4ebb-bfeb-2b35beb1d29c.
[47] *TR 139:2025 Artificial Intelligence (AI) Use Cases*, 2025, https://www.singaporestandardseshop.sg/Product/SSPdtDetail/8dfba865-c2ce-4cd3-967e-6095dc798924.
[48] Enterprise Singapore, 'SS ISO/IEC 42001:2024 Information Technology – Artificial Intelligence – Management System', Singapore Standards, 2024, https://www.singaporestandardseshop.sg/Product/SSPdtDetail/8925f190-9b21-4af4-b52c-268f7df47727.
[49] Cheong Leong Tak and Laurence Liew, 'Why Standards Matter for AI and Big Data', March 2021, https://www.enterprisesg.gov.sg/blog/quality-and-standards/why-standards-matter-for-ai-and-big-data.